

Instructions in preparing an application

Applicants are required to use this template to submit their Flagship program. Minimum font 11pts (Arial, Calibri, Times New Roman), standard line and character spacing, and normal margins. All indicated chapters are mandatory unless otherwise indicated. You must respect the indications given in each chapter in order to adhere to the expected content. A word count is clearly indicated for each section. **If you do not adhere to the maximum amount of words, your application will not be processed.** The word count includes tables, footnotes and figure legends but excludes references. Please do not include hyperlinks. The application form needs to be submitted in a single pdf document, including the budget table and letters of intent.

We recommend that you do not send in your submission at the last minute in order to avoid technical difficulties. No submission will be accepted after the deadline, March 15th at 13:00 CET. Applications should be sent in PDF format to **Flagships@erasmusmc.nl**.

Title: Organ-on-Chip technology for disease modeling and personalized medicine

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

<i>Name*</i>	<i>Institution</i>	<i>Department</i>	<i>Email</i>	<i>Tenured (i.e. fixed) position/Tenure-track/ Other (please specify)</i>	<i>For Erasmus MC applicants: clinician/ researcher/ mix</i>	<i>Role in Flagship [Main Lead/Lead/Applicant]</i>
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Corresponding Lead: Dr. Dik C. van Gent

Planned start date of seed funding: 1-10-2022

Primary Convergence theme: *Technology Supported Transitions in Health(care)*

Secondary Convergence theme: *Fundamentals of Health & Disease*

Tertiary Convergence theme (if applicable): Choose an item.

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: Personalized medicine, cancer-on-chip, health technology assessment, microfluidics, microscopy, sensors, vasculature

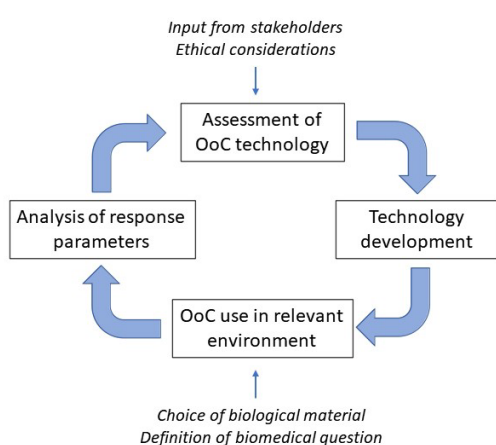
1.b Summary (max 300 words)

Please briefly describe the objectives, design, and anticipated scientific, societal and economic impact for an academic audience.

Cancer research and diagnostic techniques need improvements to optimally and efficiently enable personalized medicine in a clinical setting. Patient stratification should be refined for a better patient prognosis and to improve clinical outcome of individual patients' treatment. This will ensure that the resources spent on diagnostic techniques and treatments achieve maximum effectiveness. Recently, Organ-on-Chip (OoC) models have been introduced to realistically mimic *in vitro* normal and diseased human organs. These animal-free micro-culture systems use microfluidic technology to culture human cells and tissues under tightly controlled conditions. OoC is born out of the seamless and fruitful combination of high-tech micro/nanoscience and state-of-the-art cell biology. Hence OoC inherently requires multi-disciplinary expertise to be reliably adopted and integrated in current practice. Several challenges in (micro)engineering, tissue culturing, analysis methods, technology validation and assessment, cost effectiveness, health care organization, and ethics will have to be addressed before this technology can be used as a diagnostic clinical device.

We will create a collaborative environment with biomedical researchers and clinicians, engineers working on (microfluidic) technology, health economists, and ethicists. Together we aim to successfully develop the multi-disciplinary field of OoC and invent innovative solutions for personalized cancer treatment.

At the start of this flagship program we will investigate the needs and hurdles involved with this personalized medicine technology in the context of clinical decision making. When



technology moves forward, we will adapt the program accordingly. We will create new technological solutions (e.g. built-in sensors and devices that allow better access for microscopic imaging and complementary readout systems), improve culturing of cells and tissues (including development of a working network of blood vessels on the chip), as well as therapy response evaluation techniques. Consequently, the process of evaluation, technology development and response measurements will continue in a cyclic way (see schematic drawing).

1.c Public Summary (max 300 words)

Layman's summary.

Individual cancer patients react differently to the same therapy: a therapy that cures one patient, might not cure another patient. Therefore, developments towards so called 'personalized medicine' may enable a more effective treatment from the start, and a better outcome for the patient.

One of these developments is technology to create an 'organ-on-chip'. This mimics the tumor micro-environment, as a small piece of living tumor tissue or multicellular culture is placed into an integrated chamber which allows fluid flow. However, this technique is not yet commonly used in a medical setting, because it requires expertise from various

different disciplines. This Flagship program will integrate a unique combination of expertise in health technology assessment (EUR), tumor disease modeling (Erasmus MC), and device development allowing electro- and chemical sensing (TU Delft), forming a cancer-on-chip platform for personalized medicine.

This platform will assess safety, clinical/cost effectiveness, budget impact, ethics, legal requirements, and organizational/societal impact from bench to bedside. We will develop on 'chip' devices including a blood vessel network and optimize conditions for growing several tumor types, using breast, prostate, lung and bone tissue from patients. These devices will be made in such a way that (1) sensors can detect the reaction of the tissue; (2) culture medium with nutrients will flow through; and (3) microscopic images can be made. With the analysis of the outflow medium, we will monitor treatment responses in the tumor tissue in real time. With this collaborative approach we hope to stimulate the use of personalized medicine in the wider clinical practice and to prove that it is beneficial for patients. A positive side-effect is that a new generation of researchers, clinicians and policymakers gain insights that go beyond their own expertise.

2. Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e

- *The Flagship program contributes to the Convergence Health & Technology mission (1)*
- *The Flagship program objectives fit with the guidance per theme provided by the theme leads, and/or fit with more than one theme of Convergence Health & Technology (2)*
- *The Flagship program merges approaches and insights from distinct disciplines to reach its goals (3)*
- *The staff granted by this call actively brings together different disciplines within the program, and creates visibility and attracts talents, partners, and funding for the program (4)*
- *The Flagship program has a clear view on long term sustainability after the 5-year central funding period (5)*
- *The envisioned results of the Flagship program have the potential for a significant contribution to healthcare, society and/or have scientific impact which has the potential for long-term societal impact (6)*
- *The Flagship program provides opportunities for BSc/BA and/or MSc/MA students of the participating institutes to participate in (interdisciplinary) projects (7)*
- *The proposal defines the pathway of how this contribution can be realized. The pathway includes concrete research and/or impact pathway (societal and/or economical) valorization strategies (8)*

2.a Contribution to H&T mission and fit with H&T theme(s)

This flagship program aims to improve personalized medicine by developing technology for innovative tissue culture systems for (cancer) therapy selection. Developing technology to be used in clinical practice requires collaboration across research areas. Therefore, our aim to bring together expertise from Erasmus MC, EUR, and TU Delft to address major problems in health care aligns well with the Convergence H&T mission. Good alignment of the various stakeholder needs (i.e. of patients, clinicians, companies, members of society) is needed from an early stage to ensure that developed technologies are implemented responsibly and serve patients' and societal interests. The EUR has the necessary expertise for a thorough stakeholder analysis and early Health Technology Assessment. This will maximize the opportunities for implementation of technology in health care. Joint efforts from Erasmus MC, EUR and TU Delft will create novel solutions for better therapy response prediction to be used to maximize health output and to minimize waste of money and unnecessary patient suffering caused by ineffective treatments.

These goals will be reached by creating a collaborative environment for technology development, application to various medical problems (starting with cancer) and an open view for societal needs and challenges. This environment will create the ideal place for educating the next generation of multidisciplinary scientists, providing ample opportunity for Bachelor and Master students from various disciplines (such as engineering, nanobiology, medicine, clinical technology, and health sciences) to work together on research projects.

2.b Synergy between disciplines

This flagship program will unite seemingly distinct disciplines from the EUR, Erasmus MC and TU Delft to create a unique platform to accelerate innovative advancements in personalized medicine using cancer-on-chip as a model system.

As part of this initiative, we will carry out a comprehensive stakeholder assessment. This effort will be led by EUR, but involves all parties contributing from various disciplines. This assessment will provide insight into stakeholder needs and will help to align the engineering and biomedical technology with stakeholder needs.

Full Health Technology Assessment (HTA) requires input from all stakeholders and will be conducted continuously throughout the development of the OoC technology. Results from HTA will be used by the Erasmus MC and TU Delft to prioritize research, development of OoC technology and its implementation in clinical practice.

Close collaboration between medicine (patient materials, ethical issues) and engineering (microfluidics, readout systems, assembly), including micro/nanotechnology (organ-on-chip devices, electro- and bio-chemical sensors), biology (growing cells and tissues, as well as biochemical and microscopic analysis) places this flagship program in a unique position compared to other national and international initiatives. Several flagship participants are active partners in the 'human disease and organ model technologies' (hDMT) platform. The flagship will actively collaborate with hDMT theme groups, with a special emphasis on the Cancer-on-Chip and Vessel-on-Chip groups. In close consultation with Dr. van den Eijnden - van Raaij (hDMT director) we agreed that the projected Cancer-on-Chip facility at Erasmus MC will be a center in the national structure of Organ-on-Chip facilities that is currently evolving under supervision of hDMT. This would place the Rotterdam-Delft axis in the lead for this theme at a national level and in a very good position to take the lead in Europe and European Cancer-on-Chip initiatives. The close collaboration of the academic hospital, biomedical researchers, engineers, and health economics researchers makes this initiative highly competitive in a global environment and allows access to funding schemes that require such integration (NWA consortia in the Netherlands, EU consortia).

2.c Long term sustainability plan

Various parts of the OoC technology and applications have been developed by Erasmus MC and TU Delft (e.g. built-in sensors and a system to culture tumor slices in a microfluidic culture device), showing that the basic technology is up-and-running and that the approaches described in this application are feasible. Further implementation of OoC technology in biomedical research and personalized medicine and full exploitations of its possibilities will require further integration of efforts from the partners. Hurdles on the way to implementation in biomedical research and clinical application will be identified and addressed by this flagship program.

We aim to achieve two major technological breakthroughs: live response evaluation starting from a limited amount of biopsy material (by microscopic imaging and sensors in the microfluidic setup, and by analysis of outflow media), and creation of a functional vascular network connected to the tumor tissue to resemble the *in vivo* situation as closely as possible. These goals are ambitious and therefore the risk of failure is pronounced. However, we are convinced that we are in a very good position to address these pressing needs (felt not only by us, but also in the broader research community) by close collaboration of the engineering and biomedical partners. Experience with these subjects is present in the flagship program: advanced microscopic imaging through the Erasmus MC Optical Imaging Center (Adriaan Houtsmuller), built-in sensors through Massimo Mastrangeli and Lina Sarro, outflow analysis through John Martens and vascular biology / functional vascular networks through Caroline Cheng and Heleen van Beusekom. Even if the goals will not be reached completely (e.g. if we succeed in realizing microscopic access to the microfluidics device, but without imaging deep into 3D tissues), technological advances will be useful for many applications.

Several partners are already supported for OoC projects by external funding (e.g. KWF, NWO, NWA; total grant money over 400 k€ per year), showing the viability of attracting sufficient funding. We expect that the flagship partners will become even more attractive for funding agencies when improved facilities and collaborations in the flagship program will enhance the available infrastructure. The program will also make the technology accessible to other researchers in Rotterdam and Delft that may otherwise not have access to such facilities. Close connections to EUR-based health economics and Erasmus MC-based medical ethics makes this flagship program truly unique and improves its competitiveness.

After the five-year funding period, the facilities in Delft and Rotterdam will need to be sustained without specific flagship funding. We envision that the PI's and postdoc will acquire further funding (success already illustrated) to keep the program running. In addition, we aim to develop services and devices for Erasmus MC and TUD (and beyond) as a fee-for-service core facility. We aim to further support the flagship through development IP that could lead to sustained funding (in close collaboration with the TTO offices in Delft and Rotterdam). We do not expect major problems in this transition.

Risk:

We consider the risk of failure to be low. Our modular approach to device design and compliance with existing standards and open technology platforms will help to connect our improvements to parallel initiatives and commercial systems. We will actively follow developments in both technology (TU Delft and Erasmus MC) and stakeholder needs (EUR). Even if some technological developments would be superseded by commercial initiatives, our combination with samples from the clinic will be a unique element that is not replicated by companies.

2.d Visibility, attracting talents and new partners

The flagship program will be part of the national network of OoC centers with specific expertise, in which the Erasmus MC/TU Delft combination will specifically cover the Cancer-on-Chip expertise. This structure is supported by the Dutch OoC technology platform hDMT (see letter of support), in which several flagship PIs are actively participating in the Cancer-on-Chip working group. This platform also has a Vessel-on-Chip working group in which flagship members participate. Furthermore, European contacts will be ensured by active participation in EUROoCS, the European Organ-on-Chip Society that organizes meetings in which research from the flagship program will be presented to a wider audience. This will help to start new collaborations and give access to European funding, in addition to the national funding which is already attracted efficiently by flagship participants. These contacts will also help to attract talented PhD students and postdocs internationally.

Several flagship PIs are actively participating in teaching relevant courses, and are providing internship positions to Bachelor and Master students doing research projects from various disciplines, including Health Sciences, Medicine, Molecular Medicine, Nanobiology, Medical Ethics, Engineering, and Universities of Applied Sciences (e.g. the Biomedical Laboratory Science / BML at Hogeschool Rotterdam and Leiden). This will not only ensure proper education of the next generation of scientists, but also provide a pool of talented PhD candidates and technicians. Furthermore, it will provide opportunities to design novel teaching projects involving students from different disciplines at EUR, Erasmus MC, and TU Delft in which they can learn how to solve multidisciplinary problems.

The integrated approach to OoC technology development and implementation will also help to attract expertise, equipment, and funding from companies. At this moment we are in active contact with Micronit, Besi, Nostics, Bronkhorst, Philips, BI/OND, and Mimetas. We expect that the flagship program will significantly increase visibility and attractiveness for private partners.

2.e Long term societal impact

The flagship program mainly aims to develop, adapt and implement Organ-on-Chip technology for personalized cancer medicine. Alignment with stakeholder needs will be a continuous effort in order to prevent development of technological solutions that will not make it to the implementation phase. We envision that we will first develop approaches that can probe individual tumor sensitivity that will enable choice of the optimal therapy in the clinic.

This process will encompass several steps. First, methods have to be developed, which will include making devices and response analysis systems, as well as tissue culture methods to mimic the in vivo situation as closely as possible. Subsequently, these methods will need to be validated in a clinical setting (clinical studies to test the predictive value of the Cancer-on-Chip test), first in a proof-of-concept study and subsequently in larger prospective trials. Finally, logistics around implementation in a broader setting need to be worked out. We expect that the first five years will mainly involve development of methods and the start of proof-of-concept studies. In the next five to ten years, this can then be expanded to validated approaches, followed by implementation in a broader setting.

If this approach is successful, it may impact healthcare in several aspects. First, it will reduce the chance that treatments will not be successful. For cancer patients this will prevent many cases of progressive disease on (chemo)therapy and enable the choice for effective treatment from the start. In the area of targeted therapy, it may also have important economic impact: expensive treatments will not be started when they are not expected to be successful. Finally, it will increase quality of life for patients, as they will not be subjected to ineffective treatments that may cause severe side effects. We ensure that the necessary evidence regarding cost-effectiveness is provided throughout the development and therefore, we expect that insurance companies will start reimbursing these approaches once the clinical validity and predictive value of the tests have been established.

Important new technological breakthroughs will be patented and spin-off companies may be formed as a result of these developments. The Netherlands has an important role in European Organ-on-Chip research (e.g. EUROoCS was started recently with strong Dutch input). The flagship program would provide another boost at European level for Cancer-on-Chip approaches and could start a center of economic activity around this theme in the Rotterdam-Delft area. Most technology is currently in the development phase (TRL1-4), but could develop towards higher levels relatively quickly (we expect TRL 5 in the first five years). Several technological developments would be highly relevant for pharmaceutical industry (e.g. drug testing on a more physiologically relevant model than cell lines). Flagship partners have contacts with several companies and public parties that have expressed their support for the initiative, including Mimetas, Micronit, Biond, Bronkhorst, UPM B hDMT, and Stichting Proefdiervrij.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

- The Flagship program contributes to excellent Convergence Health & Technology research, at the forefront of international research (1)
- The Flagship program is both innovative as well as challenging in its research and approach (2)
- The Flagship program has a clear strategy to engage students in its research plan (3)
- The Flagship program has a clear aim and long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Coherence within the program is clearly demonstrated (4)
- The Flagship program aligns with or builds on (inter)national research agendas in this field, fit with the (inter)national research agendas but has a “high risk, high gain” main objective that is thoroughly explained (5)
- The Flagship program is feasible in terms of research performed, staff and budget needed (6)
- The Flagship program has a concrete plan for attracting external funding, which should result in at least the same amount as the central funding over the 5-year period (7)

3.a Ambitions

This program addresses the problem of the relatively high level of ineffective and suboptimal treatments in current oncological practice by developing novel technology for treatment response prediction. We will integrate health technology assessment (HTA), engineering solutions (for bottlenecks in microfluidics technology and microscopy), and biomedical approaches (to culture and treat tumor specimens). Innovative aspects can be found in all these areas. **First**, we are in a unique situation where HTA can shape the field from the start and interact with developers of engineering and biomedical technology. This will give guidance for research development. **Second**, we will address major challenges in the design of microfluidic setups supporting OoC devices and in readout technology.

Third, we will extend our technologies to treat (tumor) tissue specimens from the clinic in the OoC setups and correlate outcomes to clinical response in the patient.

These challenges are in line with the Convergence aims to use technology for solving medical problems (increase treatment efficacy) and make health care affordable (by preventing expensive, ineffective treatments), personalized (patient-tailored advice) and accessible (broad technology implementation by early application of HTA).

3.b Aims and feasibility

The overall **aim** of the program is to **develop OoC technology for personalized medicine**, with a special emphasis on cancer. For this purpose we define the following sub-aims:

1. Provide a clear **description of the needs and wishes of stakeholders**. This analysis will be carried out in the first year and will be updated at least once a year.
2. Provide **facilities to support OoC application** at TU Delft (development of microfluidic and analysis technology) and Erasmus MC (development of tissue culture using clinical tumor specimens). This will be set up in the first year and be extended over the five-year period.
3. **Improve tissue culture in OoC, including development of vascularization**. We will optimize conditions for growing several tumor types.
4. **Improve response analysis** by optimizing **microscopy**, built-in **sensors** and **analysis of outflow media**. For each item we expect to develop at least one working method.

Feasibility:

Most basic technology exists in the participating research groups. Expertise on HTA is available at EUR, while Erasmus MC has ample experience in assessing ethical issues. Erasmus MC developed a working OoC platform and has experience in cancer research and vascularization, while engineering capacity to adapt the system is available at TU Delft. Readout technology will build on existing expertise in microscopy, built-in sensors, and detection of biomolecules in extracellular fluids.

Staff needed:

We propose to appoint personnel for supervising facilities and providing technical support (post doc and technician), 3 PhD students to work on health technology assessment, vascularization, and microfluidics technology. The postdoc will also initiate and coordinate development of novel microscopy technology, while another postdoc (3-month assignment) will describe the ethical issues to be considered for further development of OoC technology for personalized medicine.

Bachelor and master students will be involved in all subprojects in the form of internship positions. Furthermore, we aim to incorporate teaching on OoC technology in the regular curriculum and/or specialized courses at the three institutions.

Budget:

We ask for 400 k€ per year from the convergence budget to cover personnel, consumables, travel expenses, and investment into OoC equipment.

Sustainability:

External funding is already available for both technical and biomedical projects for a total of over 500 k€ for the first 2 years. Additional funding will be attracted by the members from KWF, NWO, EU, and other sources in which they have been successful over the past years, as well as through public-private partnerships.

Possible risks:

We consider the risks to be low to moderate, as most technology is available in a basic form. The modular approach in technology development will prevent major bottlenecks (see also 2c).

3.c Work plan

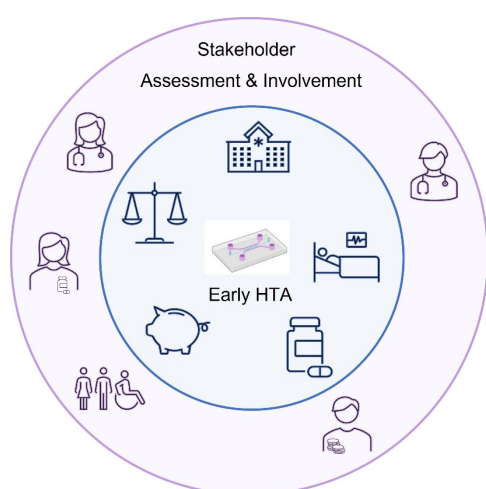
WP1: Societal and economic impact of OoC (leader: Thielen)

We will generate early Health Technology Assessment (HTA) evidence to facilitate dialogues between stakeholders and stimulate early adoption and reimbursement of OoC technology.

Task 1.1 Systematic review of theoretical frameworks

We will investigate different theoretical frameworks regarding early assessment of effectiveness, costs, and cost-effectiveness of medical devices. We will refine these to develop a novel framework for evaluation of OoC use.

Task 1.2 Comprehensive stakeholder assessment & involvement



Qualitative methods based on principles of co-creation will be employed for comprehensive stakeholder assessment including patients, clinicians, researchers, engineers, companies and citizens. This will help to define gaps and needs for which OoC technology should provide a solution. This will help to operationalize the theoretical framework and provide a basis for development of implementation strategies. The stakeholders will be involved with the program in a user committee.

Task 1.3 Early HTA

We will estimate the potential cost-effectiveness of different OoC applications from various perspectives. We will develop simulation models including predictions of costs and benefits of OoC technology for patients [1-3]. Early HTA will help prioritize research and shape technological development. Also, we will estimate the marginal value of the OoC's advantages through a so-called "headroom analysis", which enables the calculation of return on investment for OoC. Results from this task will be used to further refine OoC technology to maximize societal benefits.

Task 1.4 Simulation models Implementation strategies

We will develop decision support tools that are easy to use, based on stakeholder assessment and simulation models. These models will follow the open access scheme to allow flexible implementation and adaptation, enabling timely adoption of OoC technology.

Task 1.5 Review of ethical issues

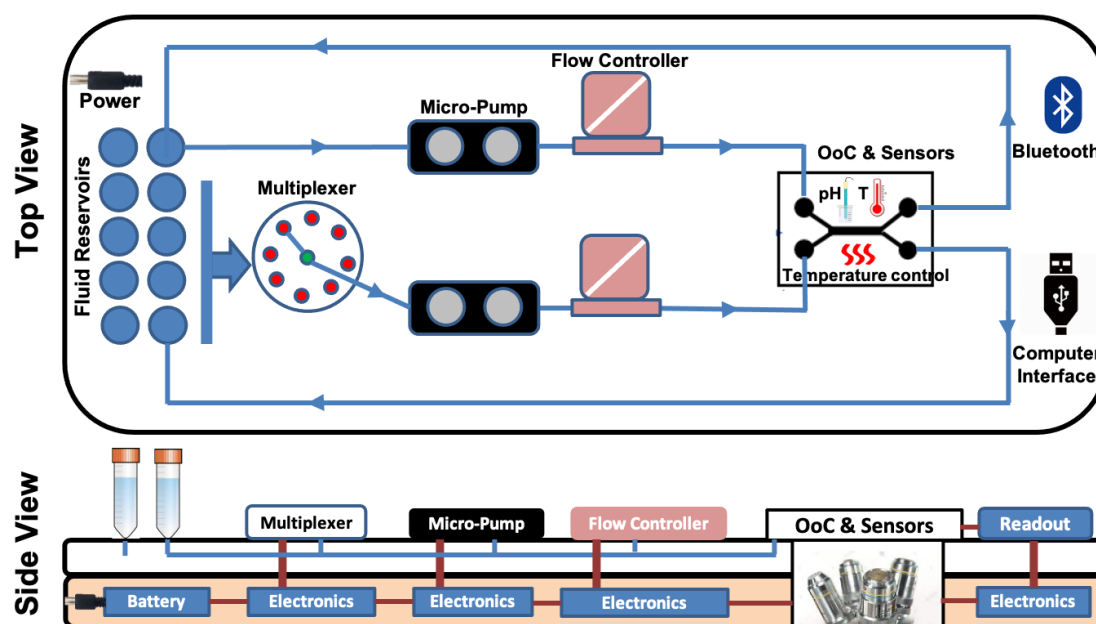
We will identify and analyze ethical issues associated with the clinical implementation of OoC technology in personalized cancer care. Adequate understanding of the ethical issues will help to anticipate and address ethical issues early in the process.

WP2: Development of OoC technology (leader: Mastrangeli)

OoC technology offers the possibility of engineering artificial (*in vitro*) microenvironments which closely mimic the *in vivo* situation by integrating microfluidic control, microfabrication, and sensing/actuation technology into single functional devices. We will address three major tasks [4].

Task 2.1 Perfusion and microfluidics

Local medium perfusion in the OoC device enhances long-term cell culturing and viability of tissues. However, current microfluidics technology cannot be integrated into a system of acceptable size. Therefore, we will develop an overall system design that takes into account ease of use, available OoC standards and platforms, control systems and software, and reusability of resources (schematic figure below).



Task 2.2 Integrated sensing

Continuous monitoring of tissue conditions within the OoC devices is addressed by developing devices that are optically transparent, to allow microscopic access and incorporation of sensors. Based on existing prototypes at TUD, we will develop technology to measure tissue confluence (trans-epithelial electrical resistance), field potential (microelectrodes), pH (electrochemical sensing), oxygen concentration, and label-free cytokine detection by surface-enhanced Raman spectroscopy.

Task 2.3 Microscopic imaging

Microscopic imaging of tissue cultures within OoC devices is a major method for evaluating tissue viability, proliferation and morphology [5]. Therefore, we will pay special attention to compatibility with microscopic imaging modalities (high-resolution, confocal, fluorescence, light sheet microscopy). Packaging material, geometrical design and

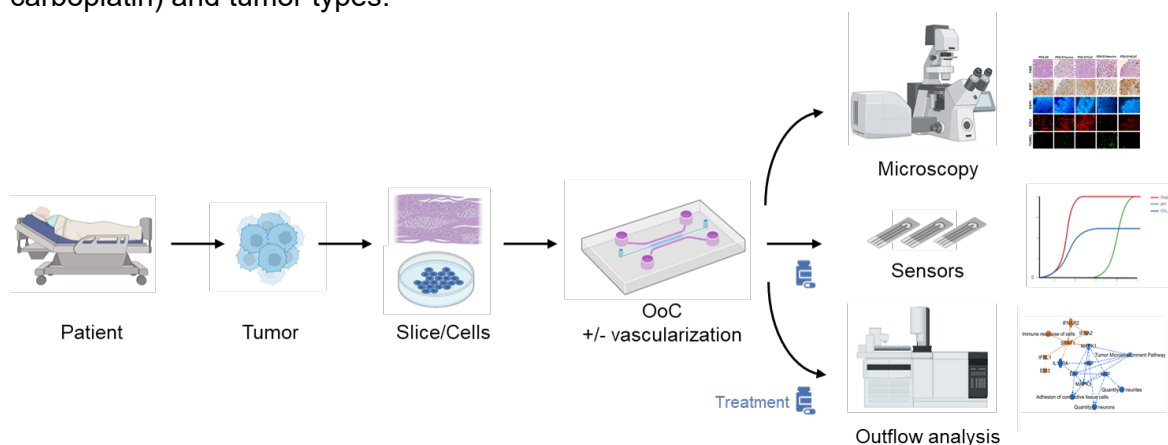
external conditioning of OoC devices will be aligned with microscopy setups for deep tissue imaging.

WP3: Human disease-on-Chip (leader: van Gent)

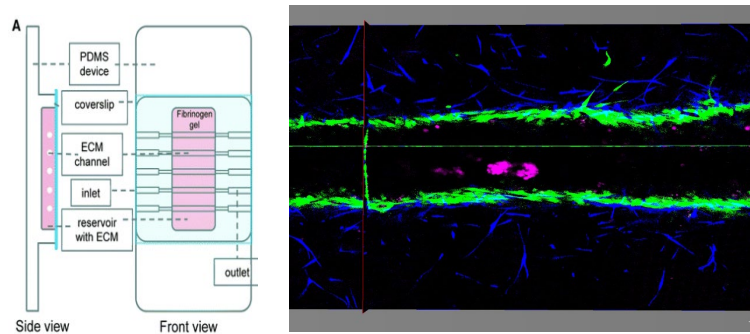
Functional drug sensitivity assays using (diseased) tissues start with culture systems that allow tissue maintenance and growth under conditions that closely resemble the *in vivo* situation. Microfluidics technologies allow precise control over growth conditions and the drugs to be delivered.

Task 3.1 Collection, growth and treatment of tumor samples

We developed methods to grow thin tumor slices in an OoC setup for more than one week [6]. We have already set up the logistics for breast cancer and showed correlation of treatment responses *in vivo* and in the OoC device for cisplatin treatment of breast tumors. We will extend this to other chemotherapy treatments (such as taxanes and carboplatin) and tumor types.



Task 3.2 Growing a vascular network in the OoC setup



(A) Current microfluidic device schematic design. (B) Flow perfusion with live imaging of circulating immune cells (THP1) in the human vessel on a chip device (GFP endothelial cells (green) and RFP mural cells (blue)).

Drugs reach tumors via the blood flow. We will therefore develop technology to grow vascular networks in the tumor tissue. We optimized cell printing of hiPSC derived vascular organoids to 3D perfused capillary beds [7]. We will combine this expertise with the optimized OoC setup to create macro-to-microfluidic human vascularized OoC technology. We will optimize coculture protocols for various tumor cell types with the vascular compartment in the OoC device.

Task 3.3 Live treatment response evaluation

We will use the technical advances in OoC design, optical imaging, built-in sensors and analysis of outflow media to monitor treatment responses in tumor specimens in real time.

Optical imaging will be done with fluorescent probes or dyes that were developed for live imaging in mice. Sensors for pH, oxygen, and biomolecules will be created or adapted. Outflow media will be monitored for DNA (indicating cell death) and its methylation status (to discriminate the various types of normal and tumor cells that this DNA originated from). We will also probe cell fate by analysis of extracellular vesicles (numbers and RNA content) in the outflow media. Finally, we will measure cytokines and other signaling molecules that are released when cells enter a senescent state. Together, these measurements will provide a readout of therapy response in time.

Task 3.4 Towards clinical validation of the predictive value of the OoC setup

The next step towards implementation of OoC technology in personalized medicine is clinical validation. We will determine the correlation between therapy response in OoC culture and in the patient. We used a similar approach to validate ex vivo treatment of tumor tissue slices and found that this was feasible [8], but that the culture system needed improvements (see section 4a). We will design similar trials when the assays have been developed to a sufficient level.

WP4: Dissemination (leaders: van Driel, Blommestein, and Ghatkesar)

Members of this program are involved as teachers and course coordinators in several bachelor and master programs. We will incorporate our research findings in our teaching. For example, in the EUR course HTA we will discuss and apply different theoretical frameworks for (early) assessment of technologies to OoC technology. We will also offer (multidisciplinary) thesis research topics in which we allow health economic, (bio)medical, and engineering students to perform their master's research in various combinations of expertise.

Furthermore, we will participate in science communication events (national and international conferences) to engage with a broad network of academia, applied universities, companies and hospitals. Our scientific publications will be open access.

WP5: Management (leaders: Thielen, Raat, and van Gent)

Task 5.1 General flagship management

The management board will handle all organizational matters, including annual flagship meetings, specialized meetings, seminars, website management and requests for admission of new research groups into the flagship. The flagship leads are responsible for coordination of annual reporting on flagship output, for which WP leaders will collect information from individual PIs. Any matters that require a change in allocation of funds or general focus of the flagship will be discussed at the annual meeting and need to be approved by the majority of flagship PIs.

Task 5.2 Technical facilities

TU Delft will extend their facilities for OoC technology development to include both fabrication and testing of devices. These facilities will be supervised by the postdoc who will handle all requests for access and discuss projects with researchers involved.

Task 5.3 Biomedical facilities

Erasmus MC will create an OoC facility with special emphasis on Cancer-on-Chip. This facility will include various microfluidic systems (both commercial systems, and custom-

made systems created in the flagship). The facility will be supervised by the postdoc, who will discuss the technical options with individual researchers and coordinate use of the OoC systems for individual projects.

3.d Alignment of the program with relevant national and international research agendas

Several flagship participants are members of hDMT theme groups (cancer-on-chip and vessel-on-chip) and activities will be coordinated with hDMT (see also letter of support). The Cancer-on-Chip facility at Erasmus MC fits in the national strategy for implementation of OoC technology and is aligned with the hDMT strategy. Flagship members are active participants in (European) OoC consortia (e.g. the ECSEL project 'Moore4Medical'). Flagship members are also members of the European Organ on Chip Society (EUROoCS) and regular participants in its meetings.

4. Consortium (25%) max 2000 words for 4.a to 4.d

- The Flagship program is driven and executed by a well-organized team of scientists from all three institutions and at least 4 departments or faculties in total. In exceptional cases, it is possible to form a Flagship program with only two leads from two institutions, but this should be clearly motivated (1)
- Members of the consortium have a track record of excellence in academic research, translation of scientific results into clinical application or societal impact (2)
- Members of the consortium have a significant track record in engaging and supervising BSc and MSc students in research projects (3)
- Members of the consortium have demonstrated to be able to attract external funding, thereby enhancing the programs sustainability after the 5 years of central funding (4)
- There is a demonstrable fit between the aims and objectives of the program and the expertise of the consortium members (5)
- End users (private and/or public sectors) are involved from an early start and team members are well-connected to external partners in the ecosystem (6)
- Governance should be clearly described and is organized in such a way that it facilitates collaboration and optimally makes use of the expertise of all participants (7)

4.a Track record of the consortium members

Track records of flagship members

Consortium members cover a broad spectrum of expertise areas, including (micro) engineering, microscopic and sensor technology, cancer biology, biochemical analysis, translational research, medical ethics, and health technology assessment. Over the past 10 years they have a track record of excellent publications in top journals (including Nature, Cancer Research, Clinical Cancer Research, Lab on a Chip). They developed technology on both the engineering side and in the biomedical realm.

Microfluidic devices have been developed (Staufer, Ghatkesar) with built-in sensors (Mastrangeli, Sarro) that form the basis for the proposed novel designs in this flagship (e.g. <https://ieeexplore.ieee.org/abstract/document/9495393> and [4, 9]). Flagship members also developed methods to grow tumor slices in OoC culture (van Gent, Jager, van Weerden) [6] and conduct response studies in primary tumor specimens and organoids [8, 10]. Other flagship members optimized a bioreactor [11] for prolonged growth of (e.g. hiPSC derived) arteries and studies of local drug delivery [12] (van Beusekom, collaboration TUD, Lifetec Group) and designed a microfluidic system for human (hiPSC derived) perfused microvessels allowing circulation and microvascular

connectivity (Cheng, collaboration TU/E; [7]). These directly relate to the cancer-on-chip cultures in the framework of this flagship.

Response analysis by direct microscopic imaging (Houtsmuller, Mezzanotte), biochemical and hypoxia measurements (Macmillan, Raat), cell-free DNA (Martens, Gribnau), and analysis of extracellular vesicles (van Driel, Martens) complements the expertise spectrum on the technical side. These researchers are all renowned experts in their own field. Adriaan Houtsmuller is head of the optical imaging center at Erasmus MC and has developed several live and super-resolution microscopy approaches [13]. Laura Mezzanotte is an expert in the area of fluorescent and luminescent imaging technology [14]. Harold Raat is a well-known hypoxia researcher, while Duncan Macmillan has ample experience in biochemical analysis. Marjolein van Driel and John Martens have experience in measurement of extracellular vesicles and cell-free DNA, while Joost Gribnau developed novel technology to measure methylation status of this DNA [15-17]. Together, these expertise areas form a broad pallet of readout technologies to be used for response evaluation.

This network of expertise is complemented by Health Technology Assessment (Thielen, Blommestein, van Baal, Uyl-de Groot), Stakeholder assessment (de Mul, Benning), and Medical Ethical embedding (Bunnik, Schermer), which will ensure optimal alignment with stakeholder needs and preferences, as well as legal and ethical considerations [1, 3, 18, 19].

Translation of scientific results into clinical application or societal impact

Consortium members have shown their ability to translate fundamental scientific discoveries into clinical applications. For example, basic concepts of homologous recombination and PARP inhibitor treatment have been translated into clinical trials (Jager, van Gent; RECAP (NL60293.078.17) and FUTURE trials (NL668566.078.18), and anthracycline sensitivity of breast tumors has been assessed in the BREAST study (Jager, van Gent; trial number NL 5588). Both studies developed technology and logistics to take biopsies from primary breast tumors or metastatic lesions, prepare them for functional assays in the research laboratory and evaluate tissue reactions using assays for cell proliferation, morphology and cell death. We envision that chemotherapy drug sensitivity assays will go through the following steps towards clinical implementation: (1) assay development, (2) validation in a proof-of-concept study, (3) prospective randomized clinical trial and (4) implementation into regular clinical practice. During the first five years we expect to concentrate on the first two steps and expand these into steps 3 and 4 over the next 5 to 10 years.

Development of technology by flagship participants has contributed to establishment of the spin-off company Biond (microfluidics technology). This flagship program is expected to increase the potential for spin-off companies considerably.

BSc and MSc student supervision and involvement

Flagship members have ample experience supervising BSc and MSc students during internships, including medicine, biomedical sciences, nanobiology, research master programs, engineering disciplines and health sciences. We expect these opportunities to increase if the flagship is implemented.

External funding

Flagship members have successfully attracted money for research projects over the past 10 years. Most PIs have 2-10 lab members funded by external projects (NWO, KWF,

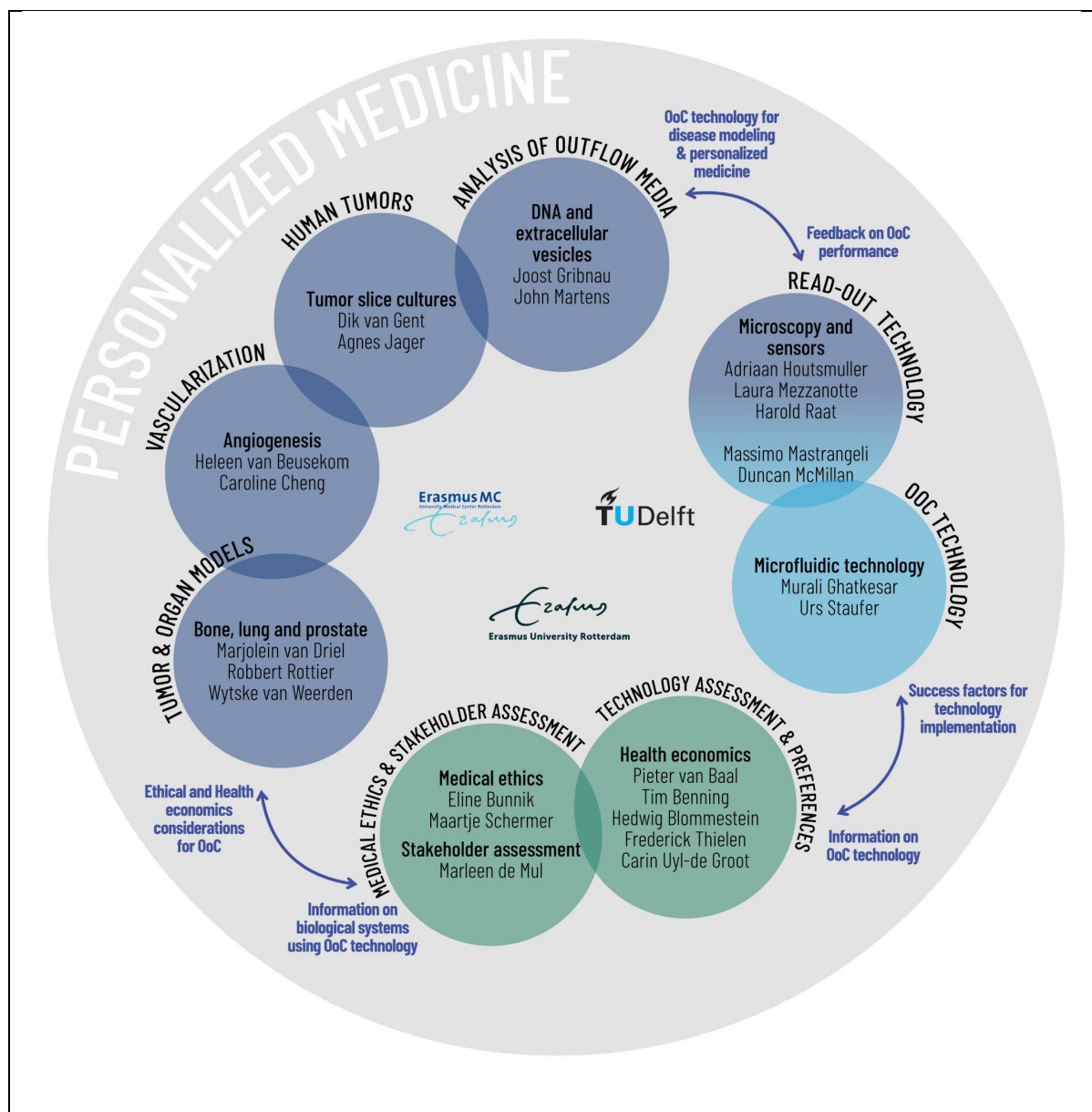
TTW, EU, public-private partnerships). Some of these projects are already in the OoC field (and will be used for matching), while other projects cover related fields. We do not expect any major difficulties to attract additional grant money for the next five years. The presence of an OoC facility and the support of the flagship will probably even increase the attractiveness of partners for funding agencies. On top of this, hDMT and Stichting Proefdiervrij have already committed themselves to support the flagship program during the 5-year period (10k€ and 25k€, respectively).

4.b Expertise of the consortium members in relation to the aims and objectives of the program

All flagship members will work towards development of OoC technology for personalized medicine, with a special emphasis on cancer, with the following specific aims:

1. Provide a clear description of the needs and wishes of stakeholders. Flagship member expertise: Frederick Thielen, Hedwig Blommestein, Pieter van Baal and Carin Uyl – de Groot – Health Technology assessment; Tim Benning – Patient Preferences; Marleen de Mul – Stakeholder Assessment / co-creation; Eline Bunnik and Maartje Schermer – Medical Ethics.
2. Provide facilities to support OoC application at TU Delft and Erasmus MC. Flagship member expertise: Murali Ghatkesar and Urs Staufer – Microfluidics Technology; Massimo Mastrangeli and Lina Sarro – Built-in Sensors; Dik van Gent and Robbert Rottier – Tissue culture in OoC; Marjolein van Driel, John Martens and Joost Gribnau – Analysis of outflow media.
3. Improve tissue culture in OoC, including development of vascularization. Flagship member expertise: Dik van Gent and Agnes Jager – Tumor slice cultures; Marjolein van Driel – bone metastasis models; Wytse van Weerden – Prostate tumor models; Caroline Cheng and Heleen van Beusekom – Vascularization.
4. Improve response analysis by optimizing microscopy, built-in sensors, and analysis of outflow media. Flagship member expertise: Adriaan Houtsmuller – Microscopy; Laura Mezzanotte – Fluorescent markers; Massimo Mastrangeli and Lina Sarro – Built-in sensors; Duncan Macmillan – Biochemical sensors; Harold Raat – Oxygen sensing; Marjolein van Driel, Joost Gribnau and John Martens – Analysis of cell-free DNA and extracellular vesicles.

The expertise of flagship members ranges from basic engineering via biochemistry, cell biology and oncology to societal aspects of implementing this new technology in clinical practice and society. We aim to strengthen the synergy between partners by regular meetings of flagship members on specific topics and at least once a year an annual flagship meeting to discuss progress and brainstorm about the issues to be addressed over the next year.



4.c Involvement of partners

The flagship program takes a cyclic approach to improving OoC technology and its use. Therefore, all partners are involved from the very start of the flagship period. We will start out by assessing various aspects of OoC technology and its application in personalized medicine for cancer treatment. The main outcome will help to choose the technologies that have the highest chance of being developed into methods to be adapted in clinical decision making and identify areas that will need special attention of regulatory bodies, adaptation in clinical practice and ethical evaluation. Ongoing improvements of OoC technology will solve major issues, such as improved options for microscopic imaging and sensing technology, which can be combined with various geometries of the OoC devices to be produced. Creation of a functional vascular network will improve interpretability and reproducibility of treatment outcomes and will move OoC technology closer to mimicking clinical behavior of the tumor under treatment. Both technological improvements will have a wider impact than only cancer biology and personalized medicine, and can therefore be considered as generic improvements of the technology that can be combined with many

different geometries of microfluidic devices. Several rounds of improvement integrating the areas described above (HTA, technological and biomedical developments) will ensure development of optimal technology that will be implemented in clinical practice.

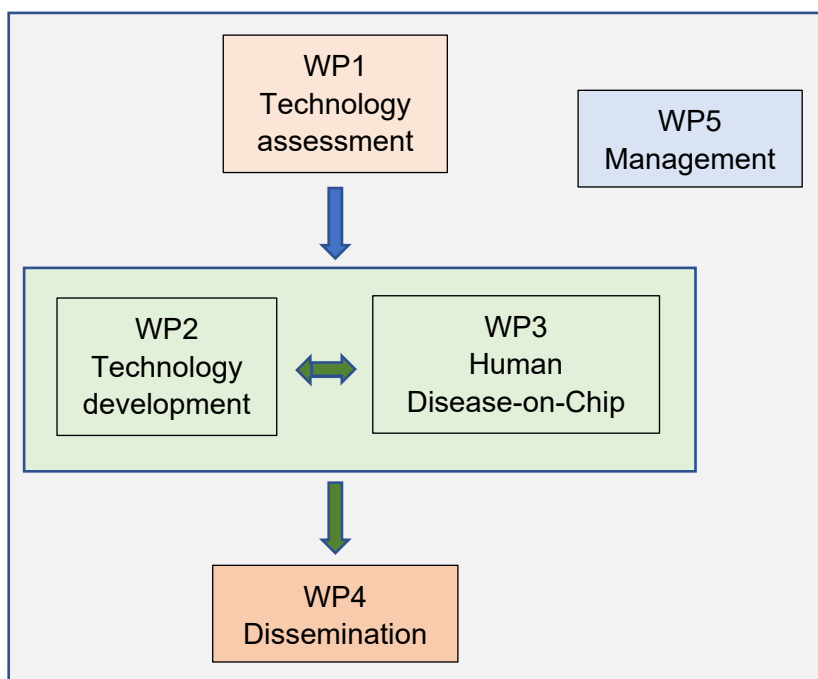
Flagship members are well connected to other researchers in the wider field of OoC technology through the hDMT platform (see also letter of support). They are active in hDMT theme groups working on Cancer-on-Chip and Vessel-on-Chip, in which they collaborate with other Dutch groups working on these subjects. Some flagship members are also members of the European Organ on Chip Society (EUROoCS) and/or regular participants in the annual EUROoCS meeting to discuss the newest technological developments with an international group of colleagues.

4.d Governance, organization and collaboration strategy of the consortium

The flagship will be managed for organizational matters by a combination of researchers from Erasmus MC (Dik van Gent, Harold Raat, Marjolein van Driel, and Robbert Rottier), EUR (Hedwig Blommestein and Frederick Thielen) and TU Delft (Massimo Mastrangeli and Murali Ghatkesar). They will discuss all matters on a regular basis (at least once a month) and also be responsible for the organization of the annual meetings. This board will also form the contact point with hDMT to discuss strategic choices that involve other national and international initiatives.

The work is subdivided into five work packages with WP leaders for each WP (see figure). WP leaders will be responsible for reporting from the WPs and for the organization of WP sessions during the annual meetings. They will also identify areas that need more specialized workshops bringing together the required experts to discuss specific topics that need to be solved to ensure progress of the WP.

We also plan to install a scientific advisory board containing several experts in the OoC area from other Dutch Universities, research facilities and companies to ensure a strong connection to the national research environment. A stakeholder committee will help to align our program with the various needs of society.



Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

Budget formulier: Convergentie Flagship Call

Flagship:
 Flagship acronym:
 Naam Pannecoeter:
 Startdatum en looptijd van het project: 1-11-2022 tot 31-10-2027
 Totaal project budget:

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Partner die de kosten maakt		Functie	Aantal te	Startdatum	Einddatum	Inhouding	Twee	Maandelijkse Kosten	2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	in cash	in kind								in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind					
Onderzoeksgeneties	Enzymia MC	proef	1.0	11/10/2022	10/10/2027	UMC - 10	5	€	4.803	€ 437.629	13.708			94.834	87.312	88.862	88.862	79.052	437.629	-	437.629			
	Enzymia MC	DO	1.0	8/10/2022	31/10/2027	UMC - 100	1	€	3.379	€ 204.287	28.187			80.480	81.260	81.260	30.750	204.287	-	204.287				
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 9	3	€	4.021	€ 141.405	5.440			34.000	35.458	35.458	37.561	31.944	181.405	-	181.405			
	Enzymia MC	proef	0.5	11/10/2022	10/10/2027	UMC - 10	5	€	4.803	€ 5.471				5.471					5.471	-	5.471			
	TU Delft	DO	1.0	8/10/2022	31/10/2027	UMC - T050	1	€	3.189	€ 230.305	28.187			57.454	57.454	57.454	58.200	28.702	230.305	-	230.305			
	EUR	DO	1.0	8/10/2022	31/10/2027	UMC - T050	1	€	3.189	€ 230.305	28.187			57.454	57.454	57.454	58.200	28.702	230.305	-	230.305			
	ICHT			0.25	11/10/2022	10/10/2027		€	444	€ 10.000										2.000	-	10.000		
	Rekening Producten			1.00	11/10/2022	10/10/2027		€	287	€ 20.000			12.500		2.000		2.000		2.000		20.000	-	20.000	
	Onderzoek 3							€	-	€ 0											-	-	-	
	Onderzoek 4								€	-	€ 0										-	-	-	
Onderzoeksgeneties	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
Onderzoeksgeneties	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
Onderzoeksgeneties	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
Onderzoeksgeneties	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
Onderzoeksgeneties	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/2027	UMC - 100	1	€	3.379	€ 100.455	16.127			60.823	60.823	60.823					€ 100.455	€ 0	100.455	
	Enzymia MC	Reassessant C	0.5	11/10/2022	10/10/1																			

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.



To: Dik van Gent
Department of Molecular Genetics
Erasmus MC
PO Box 2040
3000 CA Rotterdam
The Netherlands

Re: Letter of Support for Flagship Program Call 2021 proposal 'Organ-on-Chip technology for disease modeling and personalized medicine'

Eindhoven, 27 February 2022

Dear Dr van Gent,

With this letter we declare on behalf of the Institute for human organ and Disease Model Technologies (hDMT) our interest in and support for your research proposal 'Organ-on-Chip technology for disease modeling and personalized medicine', submitted in the Flagship Program call 2021.

As hDMT, we have read and have been fully informed about your application and we are very interested in participating in this research project.

The Institute for human organ and Disease Model Technologies (hDMT) is a consortium of renowned scientists with backgrounds in engineering, biology, physics, chemistry, pharmacology and medicine. hDMT's objective is to develop human organ and disease models that closely mimic healthy and diseased human tissues and organs, through Organ-on-Chip technology and make them available to interested users for all kind of applications by open access publication. The affiliated researchers share their knowledge, expertise and facilities to create Organ-on-Chip models, by integrating state-of-the-art human stem cell technologies and top-level engineering with a wide variety of other expertise. Currently the Consortium consists of 14 partners, including academic research institutions, university medical centers and knowledge institutes. hDMT can valorize research results through a growing network of collaborating companies.

hDMT has built a European Organ-on-Chip network of research groups in most European countries and established the European Organ-on-Chip Society (EUROoCS). EUROoCS encourages and develops Organ-on-Chip research and provides opportunities to share and advance knowledge and expertise in the field towards better health for all.

Many of the multidisciplinary researchers of the Flagship Consortium, that are involved in the development of vascularized Cancer-on-Chip models for personalized medicine, belong to the partner organizations of hDMT and are already collaborating with other scientists in different hDMT theme groups, including the Cancer-on-Chip and Vessels-on-Chip group. The proposed research fits well with hDMT's mission and with the current hDMT research priorities. Innovative Cancer-on-Chip models, that



include vascularization, mimic the *in vivo* cellular microenvironment, are transparent for live microscopic imaging and equipped with integrated sensors will be used to predict the best treatment for each individual patient. If successful, this project will result in a big step forward towards personalized medicine for cancer patients. The HTA assessment will help to further refine the Organ-on-Chip technology and realize the societal and economic benefits of the use of the new models in the clinic.

This will contribute to hDMT's dream: Imagine that once there will be Organ-on-Chip models that mimic any part of the human body, to study human sickness and health, for personalized disease treatment or even prevention, available and affordable for everyone.

Furthermore, the setup of technical and biomedical facilities focused on Cancer-on-Chip within this project aligns with the ambition of hDMT INFRA, a national infrastructure on Organ-on-Chip with Service Centers of Expertise, that aims to bring Organ-on-Chip models closer to application and is currently being set up by hDMT.

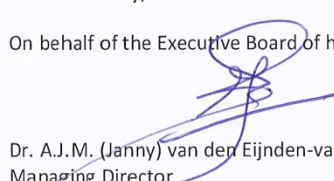
hDMT will contribute to the project by helping with the organization of scientific meetings, trainings and workshops. In addition, hDMT can enable active participation of the project researchers in the meetings, organized by hDMT and EUROoCS, and can help in disseminating the project results to the national, European, and international Organ-on-Chip community of research scientists, clinicians, regulators, patient organizations, and industry leaders.

hDMT will provide an *in-kind* contribution as itemised in the table below.

Item or activity	# of item(s) *(Cost-)price OR hours*rate	Total Amount
Co-organization of scientific meetings, trainings and workshops	50 hrs in 5 yrs (€100/hr)	5,000 euro
Dissemination of project results	50 hrs in 5 yrs (€100/hr)	5,000 euro
Total estimated value		10,000 euro

Yours sincerely,

On behalf of the Executive Board of hDMT,


Dr. A.J.M. (Janny) van den Eijnden-van Raaij
Managing Director
Tel: +31(0)6 12998074
Email: j.vandeneijnden@hdmtechnology
Website: www.hdmtechnology



Proefdienvrij zet zich in voor een toekomst zonder proefdieren. Dat doen we met hart en ziel, in de overtuiging dat de wetenschap vooruit gaat en er steeds meer technieken beschikbaar komen die onderzoek op proefdieren onnodig maken.

To: Dik van Gent
Department of Molecular Genetics
Erasmus MC
PO Box 2040
3000 CA Rotterdam
The Netherlands

March 14, 2022

Subject: Letter of Support flagship application "Organ-on-Chip technology for disease modelling and personalized medicine"

Dear Dr. Van Gent,

With this letter I declare on behalf of Stichting Proefdienvrij, our support for your flagship application "Organ-on-Chip technology for disease modelling and personalized medicine".

Proefdienvrij believes in a future without animal experiments. Every day we work hard to reach this ambitious goal. We do this by financially supporting excellent researchers and by informing a.o. the general public, scientists and students on animal free innovations. In this, we collaborate with scientist, government, and industry and take a lead in signaling, inspiring, and connecting. Supporting your application is in line with these efforts.

Our contribution to this project, if granted, is an *in-cash* contribution of 25.000 euros in 2022. We wish you all the best with the submission and selection procedure.

Yours sincerely,

Debby Weijers, MSc.
Director
Stichting Proefdienvrij

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Stichting Proefdienvrij verwerkt persoonsgegevens conform de Algemene Verordening Gegevensbescherming (AVG). Neem voor meer informatie contact met op via info@proefdienvrij.nl



To: Dik van Gent
Department of Molecular Genetics
Erasmus MC
PO Box 2040
3000 CA Rotterdam
The Netherlands

February 7, 2022

Re: Letter of Support for flagship application 'Organ-on-Chip technology for disease modeling and personalized medicine'

Dear Dik,

It is a great pleasure to write in support of your flagship application 'Organ-on-Chip technology for disease modeling and personalized medicine'.

At Mimetis, our vision is to create the simplest device for the most complex 3D biology. As true pioneers in the organ-on-a-chip biology, we develop (high-)throughput human tissue and disease models in OrganoPlates® for research and drug screening purposes. Mimetis B.V. is a privately owned biotech company based in Leiden (NL). Incorporated in 2013, we have three offices: headquarters in Leiden and subsidiaries involved in selected R&D, customer support and business development in Gaithersburg (MD) in the USA, and in Tokyo, Japan. We have assembled our various OrganoPlate® products into multiple human models, including different disease models which are relevant to your proposed project.

This initiative combines expertise from various fields, including engineering, biomedicine and health economics, in an effort to set up a comprehensive Organ-on-Chip center in the Rotterdam-Delft area. As a company in this field, Mimetis is highly interested in collaborating with this center for further development of our technologies. We intend to support the center with the technical know-how about our tissue culture systems and we are highly interested in collaborating with the flagship to validate our tissue culture models in preclinical and clinical settings. We therefore support your initiative without hesitation.

Sincerely,



Jos Joore, PhD
CEO Mimetis B.V.

**This letter constitutes a statement of good faith intentions and a recital of support to the discussions in connection with the Flagship application – "Organ-on-Chip technology for disease modeling and personalized medicine" and therefore is not intended to be legally binding upon the parties.*

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To: Dik van Gent
Department of Molecular Genetics
Erasmus MC
PO Box 2040
3000 CA Rotterdam
The Netherlands

March 10, 2022

Re: Letter of Support for flagship application 'Organ-on-Chip technology for disease modeling and personalized medicine'

Dear Dik van Gent,

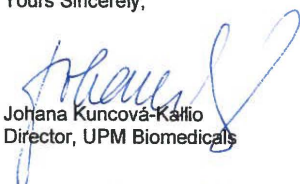
With this letter we declare our interest in and support of your flagship application 'Organ-on-Chip technology for disease modeling and personalized medicine'.

UPM Biomedicals is highly interested in collaborating setting up 'Bone metastasis (biopsy) on Chip for personalized medicine' including GrowDex hydrogel. UPM will support by providing technical know-how on GrowDex hydrogels and how they can be used in microfluidic device and with biopsy samples.

UPM Biomedicals is part of UPM – The Biofore Company that creates value from renewable and recyclable materials. UPM develops new innovative businesses, one example being high quality nanofibrillar cellulose hydrogels, which have been developed to specifically support research and new technologies in the biomedical field.

UPM Biomedicals' product GrowDex® is a biocompatible natural hydrogel that mimics the extracellular matrix (ECM) thus supporting cell growth and differentiation in three-dimensional (3D) environment. GrowDex® is made of nanofibrillar cellulose and is therefore completely xeno-free. GrowDex® has been used in numerous applications including 3D cell and organoid culture, drug release studies and 3D printing.

Yours Sincerely,



Johana Kuncová-Kallio
Director, UPM Biomedicals



Kari Luukko,
Director, Quality Assurance, UPM Biomedicals



To: Dr. D. van Gent
Department of Molecular Genetics
Erasmus MC
PO Box 2040
3000 CA ROTTERDAM
The Netherlands

Subject: Letter of Support for Flagship Program Call 2021 proposal "Organ-on-Chip technology for disease modelling and personalized medicine"

Ruurlo, March 14th, 2022

Dear Dr. Van Gent,

With this letter I declare on behalf of Bronkhorst High-Tech BV our interest in and support for your research project proposal 'Organ-on-Chip technology for disease modelling and personalized medicine' – in the Flagship Program Call 2021.

As Bronkhorst High-Tech BV, we have read and have been fully informed about your application and are very interested in participating in the project.

Bronkhorst is a worldwide organisation with over 650 employees. Our company is a leader in low flow fluidics handling technology and we offer an extensive product range of thermal, Coriolis and ultrasonic flow meters and controllers for low flow rates of gases and liquids. Our flow instruments are used for a variety of applications in laboratory, machinery, industrial and hazardous areas. By sharing our knowledge and closely cooperating with OEM customers in the field we develop customer specific low flow solutions, e.g. of multifunctional, pre-tested modules or skids for gas, liquid or vapour flow control.

Microfluidics technology is an important enabler for many applications in the life-sciences, health care and chemical / pharma industry. The ability to dose and control precise quantities of media is essential to research, as well as clinical and industrial use. An example trend is the upcoming field of Organ-On-Chip (OOC), in which organ functions and their biophysical environment are mimicked on chip-level, and processes of organ and medium interaction are studied.

OOC systems need stable and fluctuation-free fluid control, with a wide range of flow rates. Bronkhorst can support the project by providing the necessary equipment for flow and pressure measurement and control, as well as the knowledge how to apply them, as we demonstrated in a project about digestion-on-a-chip with the RUG. Moreover, since the peripheral components are quite bulky, together with the TUD and Erasmus MC we work on the development of a miniaturized flow control system for OOC applications, so, the project will also be beneficial to Bronkhorst.

Yours sincerely,
BRONKHORST HIGH-TECH B.V.


H.G. Tappel
Executive Board, Innovation

LOW FLOW FLUIDICS HANDLING TECHNOLOGY

Page 1 of 1

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III Letter of support from department head of the main lead

On headed paper of the department or institution.

Erasmus MC
University Medical Center Rotterdam

Theme Biomedical Sciences
Molecular Genetics

Direct dial +31 10 704 3199
Fax number +31 10 704 4743
Internal postal address 742
E-mail g.vanderhorst@erasmusmc.nl
Our reference GTJvdH-Je011
Date February 24, 2022

Re: Letter of Support for flagship application 'Organ-on-Chip technology for disease modeling and personalized medicine'

Dear Dik,

It is a great pleasure to write in support of your flagship application 'Organ-on-Chip technology for disease modeling and personalized medicine'.

I read the application and fully support the proposed research. I believe that it will result in important new scientific developments and that it will move forward the Convergence initiative. The department of Molecular Genetics at Erasmus MC will fully support you to act as the lead applicant on this proposal. I wish you success in the submission process and in fostering collaborations in the context of this flagship.

Sincerely yours,

Prof. dr. G.T.J. van der Horst
Head Dept. Molecular Genetics

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route
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www.erasmusmc.nl

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